



Module 1: Introduction

1. What is Artificial Intelligence (AI)?
 - a) Study of human behavior
 - b) Simulation of human intelligence in machines
 - c) Making computers slower
 - d) None of the above
2. Which of the following is NOT a foundation of AI?
 - a) Philosophy
 - b) Psychology
 - c) Sociology
 - d) Computer Science
3. Who is considered the father of AI?
 - a) Alan Turing
 - b) John McCarthy
 - c) Marvin Minsky
 - d) Arthur Samuel
4. Which of the following is an application of AI?
 - a) Natural Language Processing
 - b) Robotics
 - c) Machine Learning
 - d) All of the above
5. An intelligent agent interacts with which of the following?
 - a) Environment
 - b) Humans
 - c) Databases
 - d) None of the above
6. What is rationality in AI?
 - a) Making decisions based on logic
 - b) Always choosing the best option
 - c) Acting in a way that maximizes performance
 - d) Ignoring the environment
7. What defines the nature of environments?
 - a) Static vs. Dynamic
 - b) Deterministic vs. Stochastic



- c) Episodic vs. Sequential
- d) All of the above
- 8. Which of the following is NOT a property of a task environment?
 - a) Fully observable vs. Partially observable
 - b) Static vs. Dynamic
 - c) Symbolic vs. Numeric
 - d) Continuous vs. Discrete
- 9. Which of the following is NOT a type of agent?
 - a) Simple Reflex Agent
 - b) Model-Based Agent
 - c) Genetic Agent
 - d) Learning Agent
- 10. What is an agent program responsible for?
 - a) Interacting with the user
 - b) Controlling the agent's actions
 - c) Modifying hardware
 - d) None of the above
- 11. Which of the following best defines an agent?
 - a) A software that takes actions based on perception
 - b) A human-like robot
 - c) A system that cannot learn
 - d) A database system
- 12. What is the key characteristic of a rational agent?
 - a) It always makes correct decisions
 - b) It acts to achieve the best expected outcome
 - c) It never fails
 - d) It does not interact with its environment
- 13. Which of the following is an example of a partially observable environment?
 - a) Chess game
 - b) Poker game
 - c) Tic-Tac-Toe
 - d) Sudoku
- 14. A dynamic environment means:
 - a) The environment changes while the agent is deliberating
 - b) The environment remains static at all times
 - c) The agent changes its own state randomly
 - d) The environment is completely observable



15. What is the key advantage of a model-based agent?
 - a) It does not require memory
 - b) It can make informed decisions based on past states
 - c) It does not need to perceive the environment
 - d) It ignores uncertainty
16. What is the primary function of a learning agent?
 - a) To perform predefined actions
 - b) To improve its performance over time
 - c) To execute random tasks
 - d) To avoid interacting with the environment
17. A reflex agent operates based on:
 - a) Predefined rules and immediate perception
 - b) Complex learning algorithms
 - c) Future predictions
 - d) Stored historical data
18. What type of agent considers future actions and their outcomes?
 - a) Simple reflex agent
 - b) Goal-based agent
 - c) Reactive agent
 - d) Random agent
19. Which environment property is associated with real-time video games?
 - a) Deterministic
 - b) Episodic
 - c) Stochastic
 - d) Static
20. In AI, an agent perceives the environment using:
 - a) Sensors
 - b) Actuators
 - c) Memory
 - d) Programs



Module 2: Problem Solving

1. What is the primary objective of problem-solving in AI?
 - a) To store large amounts of data
 - b) To find a sequence of actions leading to a goal
 - c) To create random decisions
 - d) To make computers slow
2. Which search strategy explores all possible paths uniformly?
 - a) Depth-first search
 - b) Breadth-first search
 - c) Greedy search
 - d) Hill climbing
3. What is an example of an uninformed search strategy?
 - a) A* search
 - b) Breadth-first search
 - c) Hill climbing
 - d) Best-first search
4. Which of the following is an informed search strategy?
 - a) Depth-first search
 - b) Breadth-first search
 - c) A search*
 - d) Uniform cost search
5. A heuristic function is used to:
 - a) Reduce the search space in an informed search
 - b) Increase the complexity of the search
 - c) Guarantee an optimal solution
 - d) Avoid searching altogether
6. Which search strategy always expands the shallowest node first?
 - a) Depth-first search
 - b) Breadth-first search
 - c) Greedy best-first search
 - d) Iterative deepening search
7. What is the main disadvantage of depth-first search?
 - a) It may not guarantee the shortest path
 - b) It requires a lot of memory
 - c) It is always optimal
 - d) It never terminates



8. What is the primary goal of iterative deepening search?
 - a) To combine the benefits of depth-first and breadth-first search
 - b) To improve memory usage
 - c) To avoid using heuristics
 - d) To randomly explore the search space
9. *In A search, the evaluation function $f(n)$ is given by:**
 - a) $g(n) + h(n)$
 - b) $g(n) - h(n)$
 - c) $g(n) * h(n)$
 - d) $g(n) / h(n)$
10. What is the purpose of a heuristic function in informed search strategies?
 - a) To estimate the cost from the current node to the goal
 - b) To reduce computational complexity
 - c) To guarantee an optimal solution
 - d) To avoid searching
11. What is the Minimax algorithm used for?
 - a) Image processing
 - b) Game playing
 - c) Neural networks
 - d) Data mining
12. What does Alpha-Beta pruning improve in Minimax?
 - a) Memory storage
 - b) Computational efficiency
 - c) Image processing speed
 - d) None of the above
13. In adversarial search, what does a game tree represent?
 - a) The structure of a neural network
 - b) All possible moves in a game
 - c) A clustering technique
 - d) A type of decision tree
14. What is the primary objective of the Minimax algorithm?
 - a) To maximize the number of moves in a game
 - b) To minimize computation time
 - c) To find the best possible move assuming an optimal opponent
 - d) To randomly select moves
15. Which search strategy is commonly used in two-player games?
 - a) Breadth-first search
 - b) A* search



- c) Minimax search
- d) Best-first search
- 16. What does Alpha-Beta pruning do in the Minimax algorithm?
 - a) Eliminates branches that do not influence the final decision
 - b) Ensures an optimal solution
 - c) Reduces the branching factor
 - d) Increases the depth
- 17. What is a Constraint Satisfaction Problem (CSP)?
 - a) A problem defined by constraints on variables
 - b) A problem that has no solution
 - c) A problem solved by brute force
 - d) A problem allowing only one solution
- 18. Which of the following is an example of a CSP?
 - a) Chess
 - b) Sudoku
 - c) Pathfinding
 - d) Sorting
- 19. What is constraint propagation in CSPs?
 - a) A technique to deduce new constraints from existing ones
 - b) A method for increasing computation speed
 - c) A way to randomly generate solutions
 - d) A technique to eliminate search
- 20. What is the main advantage of backtracking in CSPs?
 - a) It guarantees an optimal solution
 - b) It explores all possible solutions efficiently
 - c) It systematically eliminates invalid solutions
 - d) It reduces memory usage



Module 3: Search in Complex Environments

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 - a) Image processing
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10. What is the main advantage of backtracking in CSPs?
 - a) It guarantees an optimal solution
 - b) It explores all possible solutions efficiently
 - c) It systematically eliminates invalid solutions
 - d) It reduces memory usage
11. What type of environment does real-time strategy games belong to?
 - a) Deterministic
 - b) Stochastic
 - c) Episodic
 - d) Static
12. What is hill climbing used for in search?
 - a) Exploring all possible states
 - b) Finding solutions by evaluating local states
 - c) Reducing branching factors
 - d) Minimizing memory usage
13. What is local search commonly applied to?
 - a) Exact solutions
 - b) Optimization problems
 - c) Tree structures
 - d) Pathfinding
14. What is simulated annealing inspired by?
 - a) The process of cooling metals
 - b) Biological evolution
 - c) Neural connections
 - d) Graph theory
15. What is a plateau problem in local search?
 - a) When the solution is not unique
 - b) When multiple neighboring states have equal evaluation



- c) When the heuristic is unavailable
- d) When the system crashes
- 16. What is a fitness function used for in evolutionary algorithms?
 - a) Selecting random solutions
 - b) Generating initial populations
 - c) Evaluating the quality of solutions
 - d) Defining data types
- 17. Which of the following represents crossover in genetic algorithms?
 - a) Combining two parent solutions to produce offspring
 - b) Selecting the best solution
 - c) Mutating data randomly
 - d) Terminating iterations
- 18. What is mutation in genetic algorithms?
 - a) Randomly altering parts of a solution to increase diversity
 - b) Reproducing solutions
 - c) Terminating search
 - d) Evaluating solutions
- 19. What is a goal test in search algorithms?
 - a) A condition to check whether the goal state has been reached
 - b) A method to find alternative paths
 - c) A technique to estimate solution quality
 - d) A brute-force technique
- 20. What does "elitism" refer to in genetic algorithms?
 - a) Preserving the best individuals in the next generation
 - b) Discarding poor solutions
 - c) Using no mutation
 - d) Reducing the size of the population



Module 4: Knowledge Representation and Reasoning

1. Which of the following is a knowledge representation technique?
 - a) Decision Trees
 - b) First Order Logic
 - c) Neural Networks
 - d) Genetic Algorithms
2. What does propositional logic deal with?
 - a) Statements that are true or false
 - b) Objects and relations
 - c) Probability distributions
 - d) Fuzzy reasoning
3. What is First Order Logic (FOL) also called?
 - a) Predicate logic
 - b) Propositional calculus
 - c) Decision logic
 - d) Machine logic
4. What does a knowledge base contain in AI?
 - a) Random facts
 - b) Facts and rules for reasoning
 - c) Only solutions
 - d) Non-verifiable data
5. Which of the following is an inference technique?
 - a) Modus Ponens
 - b) Forward chaining
 - c) Backpropagation
 - d) Both a and b
6. What does the term "semantic network" refer to?
 - a) A type of neural network
 - b) A graph-based structure for representing concepts
 - c) A search algorithm
 - d) An AI model for learning
7. What is the frame problem in AI?
 - a) Representing images
 - b) Overfitting models
 - c) Updating facts when an agent performs actions
 - d) Avoiding recursive rules



8. Which reasoning method handles uncertain information?
 - a) Deductive reasoning
 - b) Probabilistic reasoning
 - c) Model-based reasoning
 - d) Constraint-based reasoning
9. What is the difference between declarative and procedural knowledge?
 - a) Declarative is about actions; procedural is about facts
 - b) Declarative is about facts; procedural is about actions
 - c) Both are about rules
 - d) Both involve deduction
10. Which AI system simulates human reasoning using rules?
 - a) Decision trees
 - b) Expert systems
 - c) Genetic algorithms
 - d) Neural networks
11. What is a knowledge graph?
 - a) A decision-making system
 - b) A visual representation of neural pathways
 - c) A structured representation of real-world entities and relationships
 - d) A fuzzy reasoning model
12. What does ontology represent in AI?
 - a) Theoretical frameworks for ML
 - b) Relationships between data and concepts
 - c) Heuristic evaluation strategies
 - d) Output predictions
13. Which reasoning method is based on fuzzy sets?
 - a) Symbolic reasoning
 - b) Fuzzy logic
 - c) Inductive reasoning
 - d) Deductive reasoning
14. What is backward chaining used for?
 - a) Proving the most general rules
 - b) Starting with goals and finding supporting facts
 - c) Exploring all facts sequentially
 - d) Avoiding loops
15. Which AI approach relies on if-then rules for decision-making?
 - a) Genetic algorithms
 - b) Production systems



- c) Clustering models
- d) Deep learning
- 16. What is semantic reasoning?
 - a) Solving puzzles in symbolic logic
 - b) Reasoning based on language semantics
 - c) Performing reinforcement learning
 - d) Evaluating complex search nodes
- 17. Which representation method uses axioms to define relationships?
 - a) Frames
 - b) Logic-based representation
 - c) Artificial neural networks
 - d) Genetic programming
- 18. What is a major problem with rule-based systems?
 - a) They always produce wrong results
 - b) They require constant training
 - c) They cannot handle incomplete knowledge
 - d) They cannot be automated
- 19. What is a frame in AI?
 - a) A method for backward chaining
 - b) A template for representing stereotypical situations
 - c) A logic gate
 - d) A constraint in decision trees
- 20. What does default reasoning handle?
 - a) Real-time data
 - b) Exceptions to general rules
 - c) Probabilistic outcomes
 - d) Image processing tasks



Module 5: Machine Learning

1. What is the main goal of supervised learning?
 - a) To learn patterns from labeled data
 - b) To learn without any supervision
 - c) To generate random outputs
 - d) To slow down computation
2. Which algorithm is commonly used for classification?
 - a) K-Nearest Neighbors
 - b) A* Algorithm
 - c) Dijkstra's Algorithm
 - d) Quick Sort
3. Which of the following is an example of unsupervised learning?
 - a) Regression
 - b) Clustering
 - c) Classification
 - d) Reinforcement
4. What does reinforcement learning involve?
 - a) Learning by interacting with an environment and receiving feedback
 - b) Pre-labeled data
 - c) Grouping similar items
 - d) Static rules
5. What is overfitting?
 - a) When the model performs poorly
 - b) When the model performs well on training data but poorly on unseen data
 - c) When the data size is insufficient
 - d) When the model uses too few features
6. What is a decision tree?
 - a) A model that makes decisions based on a tree-like structure
 - b) A neural network
 - c) A clustering algorithm
 - d) A search method
7. Which algorithm is commonly used for regression?
 - a) Linear regression
 - b) K-means clustering
 - c) Decision trees
 - d) Naive Bayes



8. What is cross-validation used for?
 - a) Evaluating model performance on unseen data
 - b) Selecting hyperparameters
 - c) Improving performance
 - d) Making predictions
9. What is a neural network inspired by?
 - a) The human brain
 - b) Evolution
 - c) Genetic mutation
 - d) Fuzzy logic
10. What does backpropagation do in neural networks?
 - a) Adjusts weights based on errors
 - b) Optimizes feature selection
 - c) Prunes decision trees
 - d) Combines clustering outputs
11. What is a support vector machine used for?
 - a) Regression tasks
 - b) Data sorting
 - c) Classification and regression tasks
 - d) Clustering
12. Which learning method uses labeled and unlabeled data?
 - a) Unsupervised learning
 - b) Reinforcement learning
 - c) Semi-supervised learning
 - d) Supervised learning
13. What is a confusion matrix used for?
 - a) Measuring memory usage
 - b) Evaluating classification models
 - c) Visualizing datasets
 - d) Optimizing hyperparameters
14. Which machine learning model uses ensemble learning?
 - a) Random Forest
 - b) K-means
 - c) Linear Regression
 - d) Gradient Descent
15. What is the objective of gradient descent?
 - a) Minimize loss function
 - b) Maximize data points



- c) Randomize weights
- d) Predict probabilities
- 16. Which metric is suitable for imbalanced classification problems?
 - a) Accuracy
 - b) ROC-AUC
 - c) Mean squared error
 - d) Entropy
- 17. What is feature scaling used for?
 - a) Balancing features on different scales
 - b) Data aggregation
 - c) Eliminating redundant features
 - d) Data compression
- 18. What is hyperparameter tuning?
 - a) Optimizing model parameters during training
 - b) Adjusting predefined parameters to improve performance
 - c) Automating feature selection
 - d) Simplifying neural networks
- 19. What does "Bagging" in machine learning refer to?
 - a) Combining predictions of multiple models
 - b) Stopping overfitting
 - c) Reducing computation
 - d) Data compression
- 20. Which optimization method is often faster than batch gradient descent?
 - a) Mini-batch gradient descent
 - b) Genetic algorithms
 - c) Reinforcement learning
 - d) Naive Bayes



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Answer Key

Question No.	Module 1	Module 2	Module 3	Module 4	Module 5
1	b	a	b	b	a
2	c	b	b	b	b
3	b	b	b	c	a
4	d	c	c	b	c
5	a	a	a	d	b
6	c	b	a	a	a
7	d	b	b	b	b
8	c	a	b	a	c
9	c	a	b	c	b
10	b	a	c	b	a
11	a	b	b	a	c
12	b	b	a	b	a
13	b	a	c	b	a
14	a	a	b	b	c
15	b	a	c	a	c
16	b	a	b	b	b
17	a	c	b	a	a
18	b	a	b	c	b
19	c	b	c	b	a
20	a	a	c	a	b